

Day-04

Python interview questions on conditional statements:

Basic Questions:

- 1.What are conditional statements in Python?
- 2.Explain the difference between if, elif, and else.
- 3.How does Python handle multiple conditions in a single if statement?
- 4.What happens if you write an if condition without an else statement?
- 5.How do you write a one-liner if statement in Python?

Intermediate Questions:

- 1.What is the difference between if x: and if x == True:?
- 2.How do you use logical operators (and, or, not) in conditional statements?
- 3.What will be the output of this code?

code

```
x = 5
```

```
if x > 3 and x < 10:
```

```
    print("Valid")
```

- 4.How does Python evaluate expressions with multiple conditions?
- 5.What is short-circuiting in Python conditional statements?

Advanced Questions:

- 1.Can you use a conditional statement inside a list comprehension? Provide an example.
- 2.What is the difference between a ternary operator and a normal if-else statement?
- 3.Explain how match-case (introduced in Python 3.10) is different from if-elif-else.
- 4.What will be the output of the following nested if condition?

code

```
x = 7
```

```
if x > 5:
```

```
    if x < 10:
```

```
        print("Between 5 and 10")
```

```
    else:
```

```
        print("Greater than 10")
```

5.How can you avoid deep nesting in conditional statements?